PAEDIATRIC ORTHO · NEET-SS / FET / INI-SS

clubfoot and foot deformities

Congenital talipes equinovarus (clubfoot)

- Deformity **CAVE**: **C**avus (plantarflexed first ray), forefoot **A**dductus, hindfoot **V**arus, **E**quinus.
- The talar neck is deviated medially and plantarward; the navicular is displaced medially. Twice as common in boys; half are bilateral.
- **Pirani score** (0-6: 3 midfoot + 3 hindfoot) and Dimeglio grade assess severity and track correction.

Ponseti method

1. Correct the **cavus first** by **supinating the forefoot** (lift the first metatarsal) to line it up with the hindfoot.
2. **Abduct** the supinated foot around the **head of the talus** as fulcrum. Never press on the calcaneocuboid joint (**Kite's error**), and never pronate.
3. Weekly long-leg casts, usually 5-6.
4. **Percutaneous Achilles tenotomy** (needed in about 80-90%) once abduction of about 60-70 degrees is reached but ankle dorsiflexion stays under about 10-15 degrees; then 3 weeks in a cast.
5. **Foot abduction brace**: 70 degrees external rotation on the clubfoot side (about 40 on a normal side), 10-15 degrees dorsiflexion; 23 hours a day for 3 months, then nights and naps until 4-5 years.

- **Relapse** is mostly due to stopping the brace. Re-cast. Dynamic supination in a child over about 3 years: **tibialis anterior transfer to the lateral cuneiform**.

- **Complex (atypical) clubfoot:** short, fat foot, deep plantar crease, short hyperextended big toe. Use the modified Ponseti technique.

Other foot deformities

- **Congenital vertical talus (rocker-bottom):** irreducible dorsal dislocation of the navicular on the talus. **Forced plantarflexion lateral view:** the talar axis still passes plantar to the first metatarsal and cuneiform (**oblique talus** reduces). Treat with **reverse Ponseti (Dobbs)** casting, then limited open pinning of the talonavicular joint and Achilles tenotomy. Associated with arthrogryposis, myelomeningocele and chromosomal syndromes.
- **Metatarsus adductus:** a kidney-shaped foot with a flexible heel; the **heel bisector** (Bleck) grades it. Most resolve; stretch or cast the rigid ones.
- **Tarsal coalition** (peroneal spastic flatfoot, recurrent ankle sprains):
  - **Calcaneonavicular:** age 8-12, **"anteater" sign** on the oblique view. Excise and interpose EDB or fat.
  - **Talocalcaneal (middle facet):** age 12-16, **C-sign** on the lateral view; CT is the test of choice. Resect (with fat interposition) if the coalition is under about 50% of the subtalar joint surface and there is no arthritis; otherwise fuse.
- **Flexible flatfoot** is physiological if the arch appears on tiptoe and the hindfoot swings into varus. Treat only if painful.
- **Cavovarus foot in a child:** look for a neurological cause (Charcot-Marie-Tooth, spinal dysraphism). MRI of the spine if unilateral or progressive.

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.